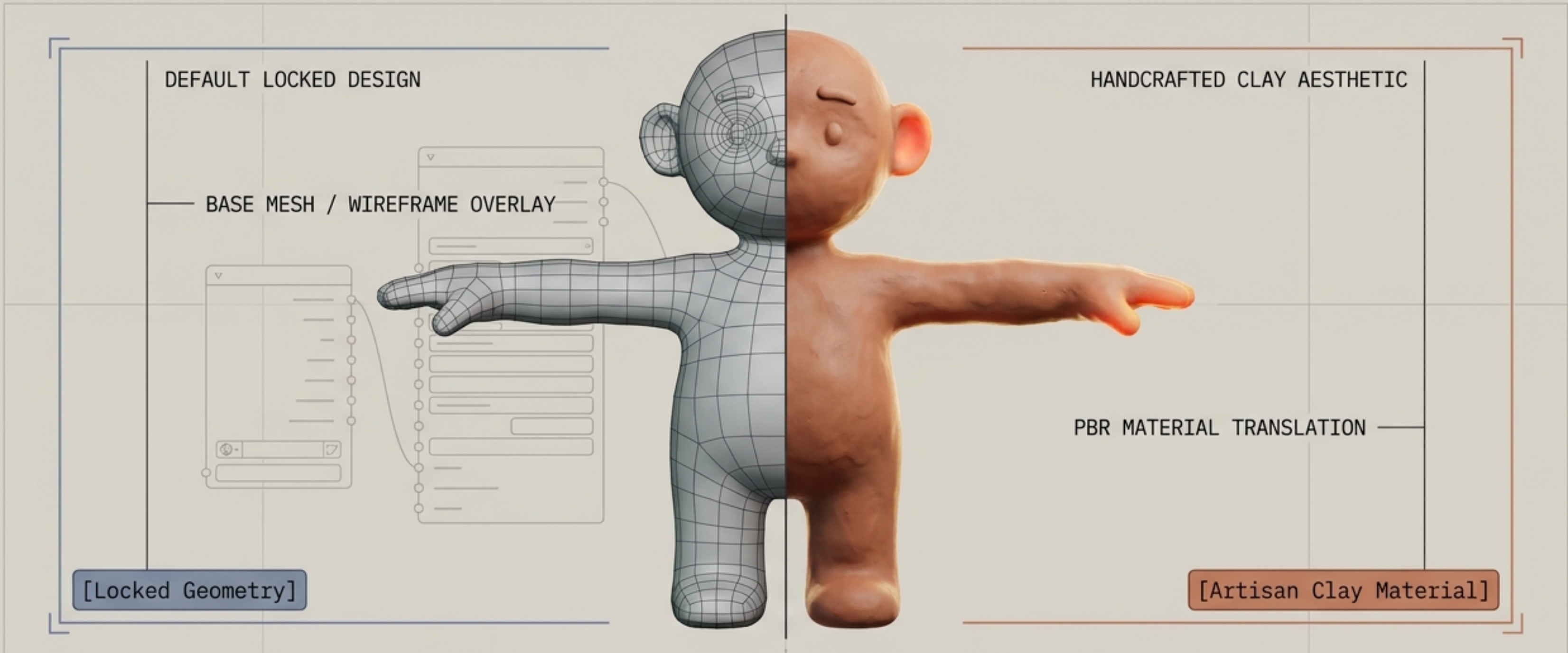


KFB × ClayBound Production Specification

Handcrafted PBR Translation System for Blender 4.x



[Soft Volumes]

[Velvet Matte Finish]

[Subsurface Warmth]

[Animation-Safe]

The Digital Sculptor's Spec Sheet

Macro Form

Rounded silhouettes,
complete absence of sharp
90° polygonal edges.

[OBSERVED]

Meso Detail

Subtle tool indentations
and soft spatial
compression.

[INFERRED]

Micro Surface

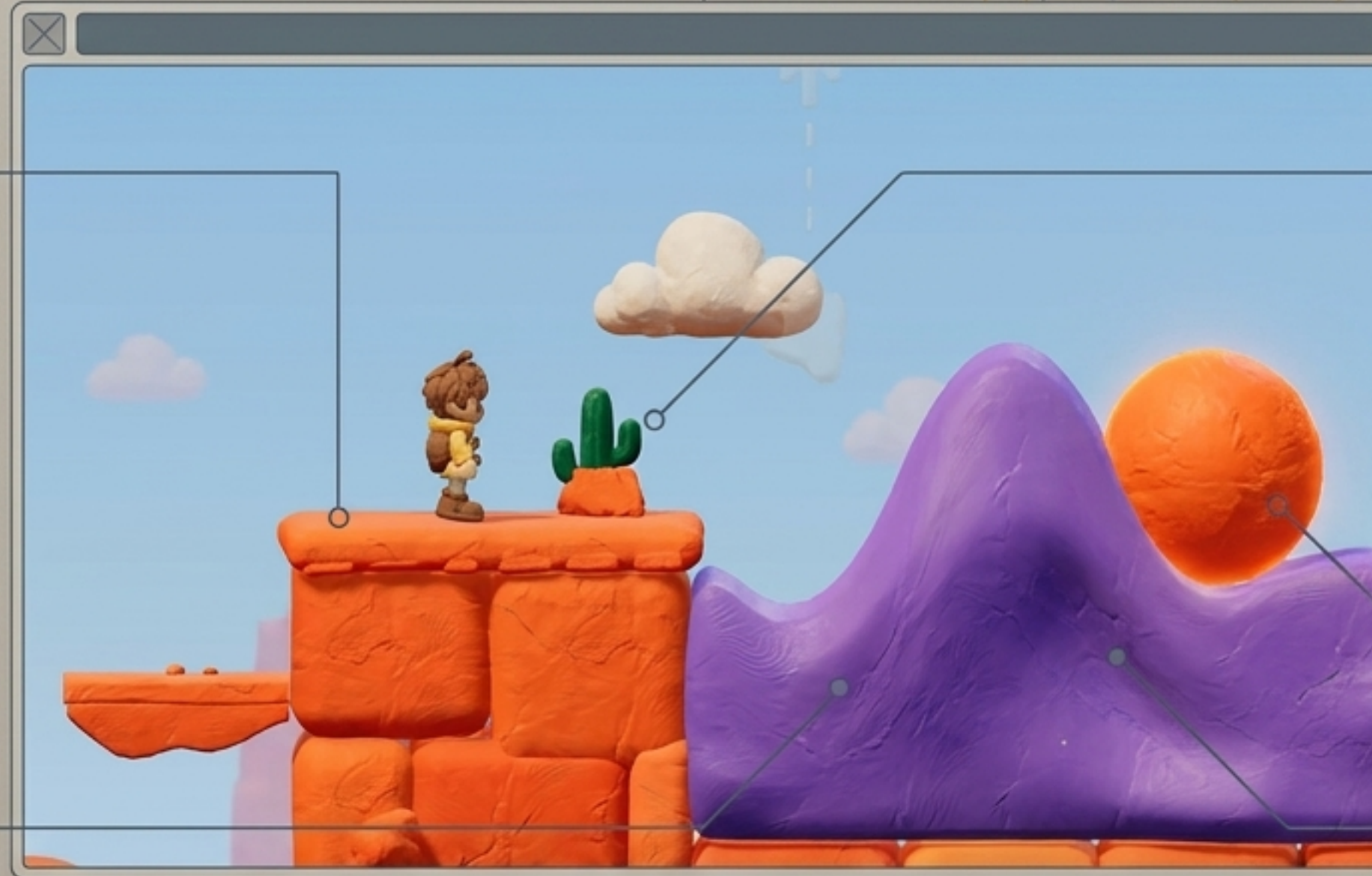
Starch-like porous grain,
strictly dielectric, zero
glossy clearcoat.

[DOCUMENTED]

Light Transport

Red-shifted subsurface
scattering visible at thin
extremities.

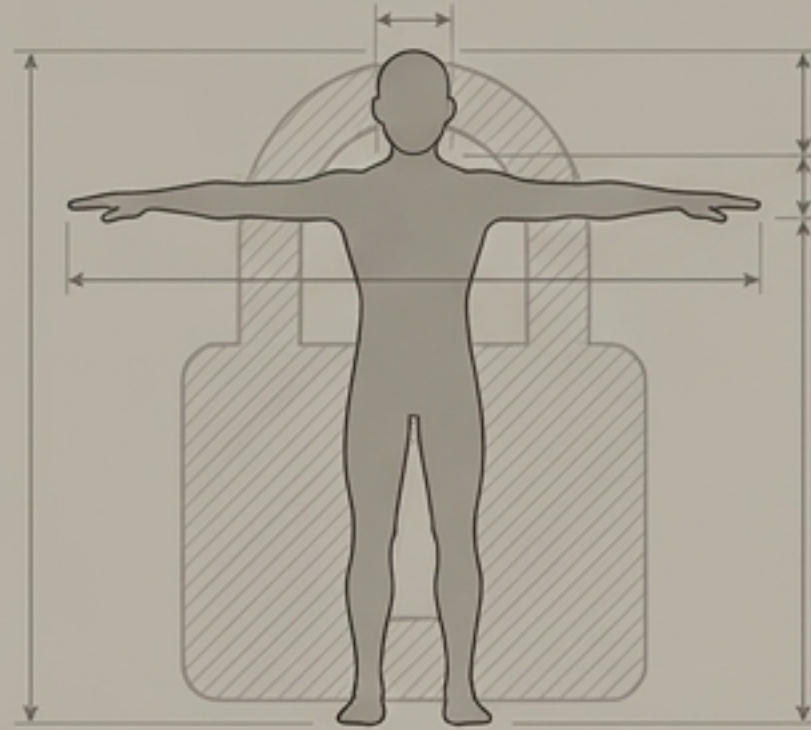
[OBSERVED]



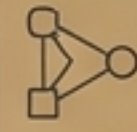
The Digital Sculptor's Spec Sheet



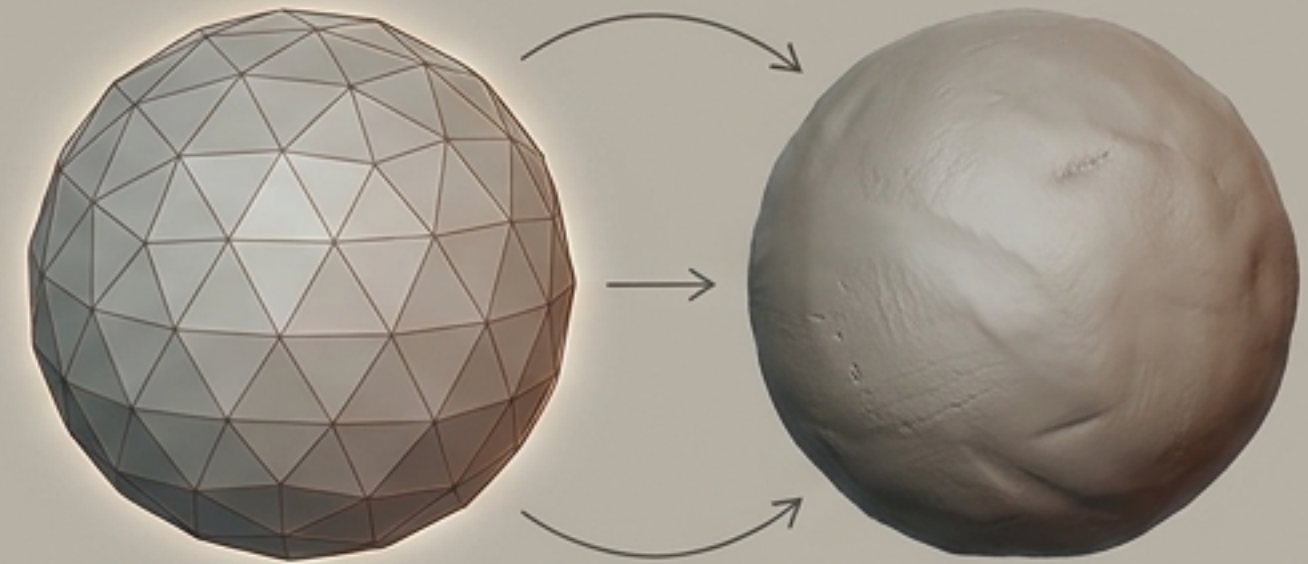
LOCKED (Preserve KFB Identity)



- Base Geometry proportions and topology outline
- Distinctive silhouettes and facial feature spacing
- Rigging weights and skeletal structure
- Core base color palettes



UNLOCKED (Target for Modification)



- Edge topology (non-destructive via Modifiers only)
- Surface normal evaluation
- PBR material responses (Roughness, SSS)
- Micro-displacement and bump mapping
- Environmental lighting setup

The Digital Sculptor's Spec Sheet

Tier 3: Textures (Micro Scale)

Mechanism: Image Maps

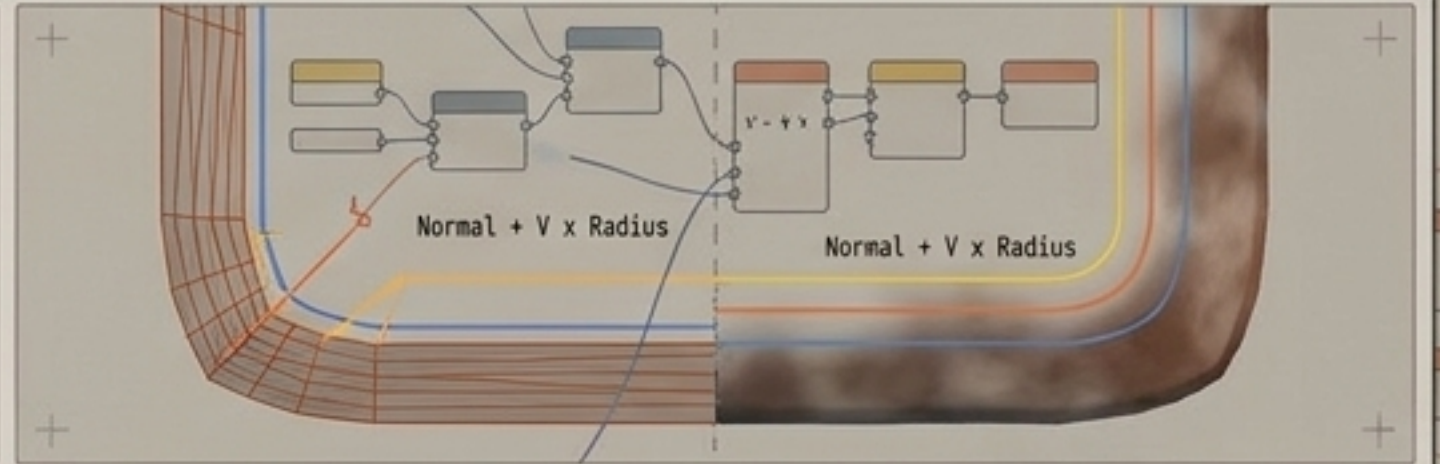
Parameters:
Tangent-space high-frequency
Normal maps for
organic porosity.



Tier 2: Shader (Meso Scale)

Mechanism: Vector Math

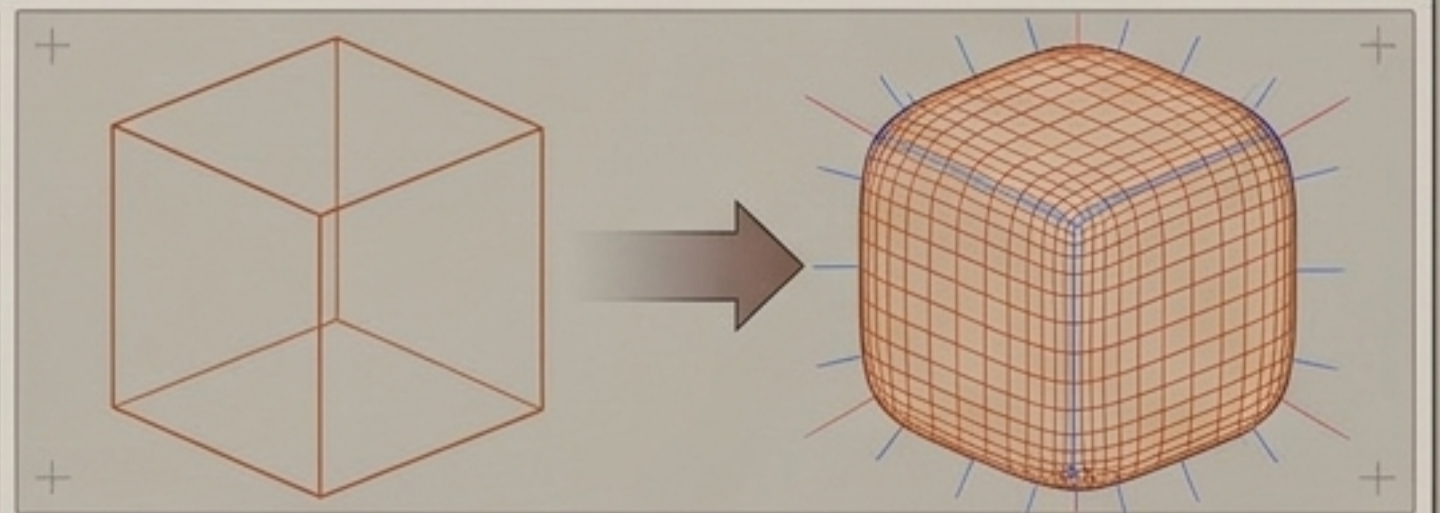
Parameters:
Bevel Shader Node
(Radius: 0.010m) for
interior crevice rounding
+ Mottle Mask for broad
color breakup.



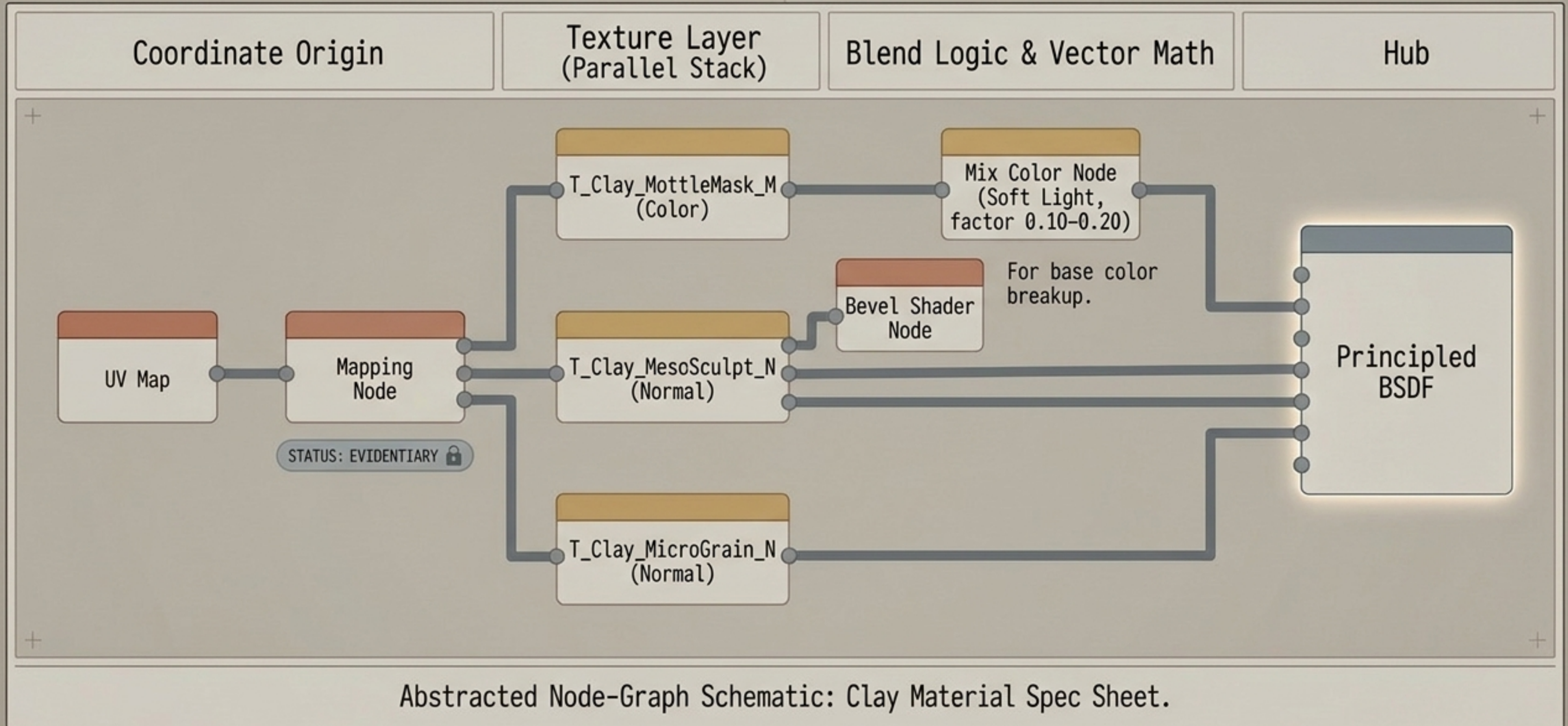
Tier 1: Geometry (Macro Scale)

Mechanism: Modifiers

Parameters:
Subdivision Surface (Lvl
1-2) + Bevel Modifier
(Width: 0.005m, Angle
limit: 30°) + Weighted
Normals.



The Digital Sculptor's Spec Sheet



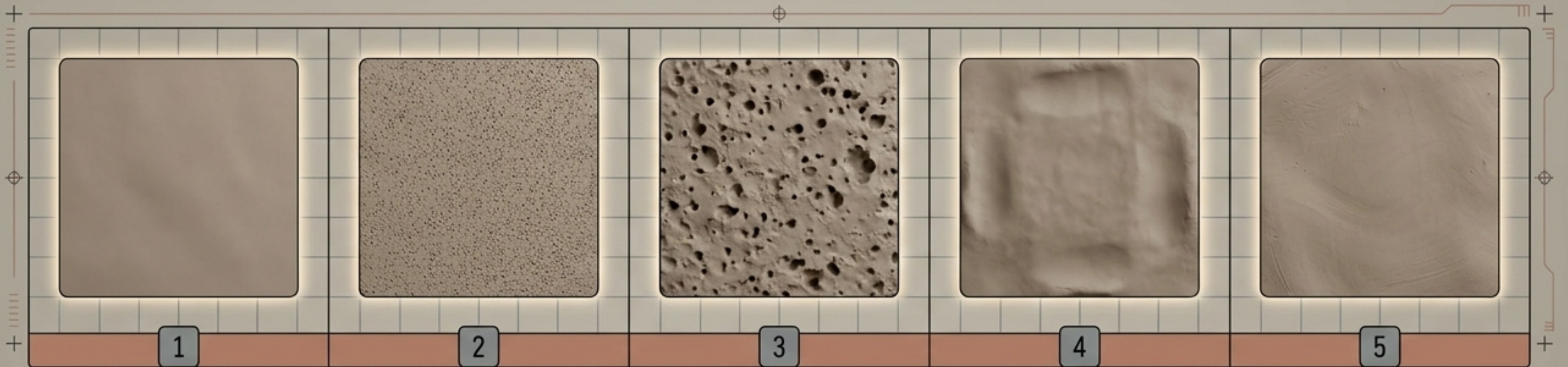
The Digital Sculptor's Spec Sheet

Parameter	Value / Rationale
Roughness	0.65 - 0.85 (Matte finish, broad tool-mark highlights)
Metallic / Coat	0.000 (Strictly dielectric, avoids plastic glaze look)
IOR	1.450 (Organic plasticine index)
Specular IOR Level	0.450 (Dry tactile appearance)
Subsurface Weight	0.250 (Internal translucency without waxy jelly look)
Subsurface Radius	R: 1.000, G: 0.350, B: 0.200 (Warm red-shifted internal glow)
Subsurface Scale	0.020m (Relative to a 1.8m standardized asset)
Sheen Weight	0.150 (Micro-fiber edge softening)

PHASE 1 — CORE CLAY QUEUE (Sequential Generation Required)

Universal Rules

Perfect X/Y seamless tiling. No frames. No central hero objects.
No baked lighting or cast shadows. No procedural fingerprint cliché.



1. Smooth Matte Clay:
The primary base
standard.
(START HERE)

2. Fine-Grain Clay:
High-frequency
micro-porosity
(starch-like).

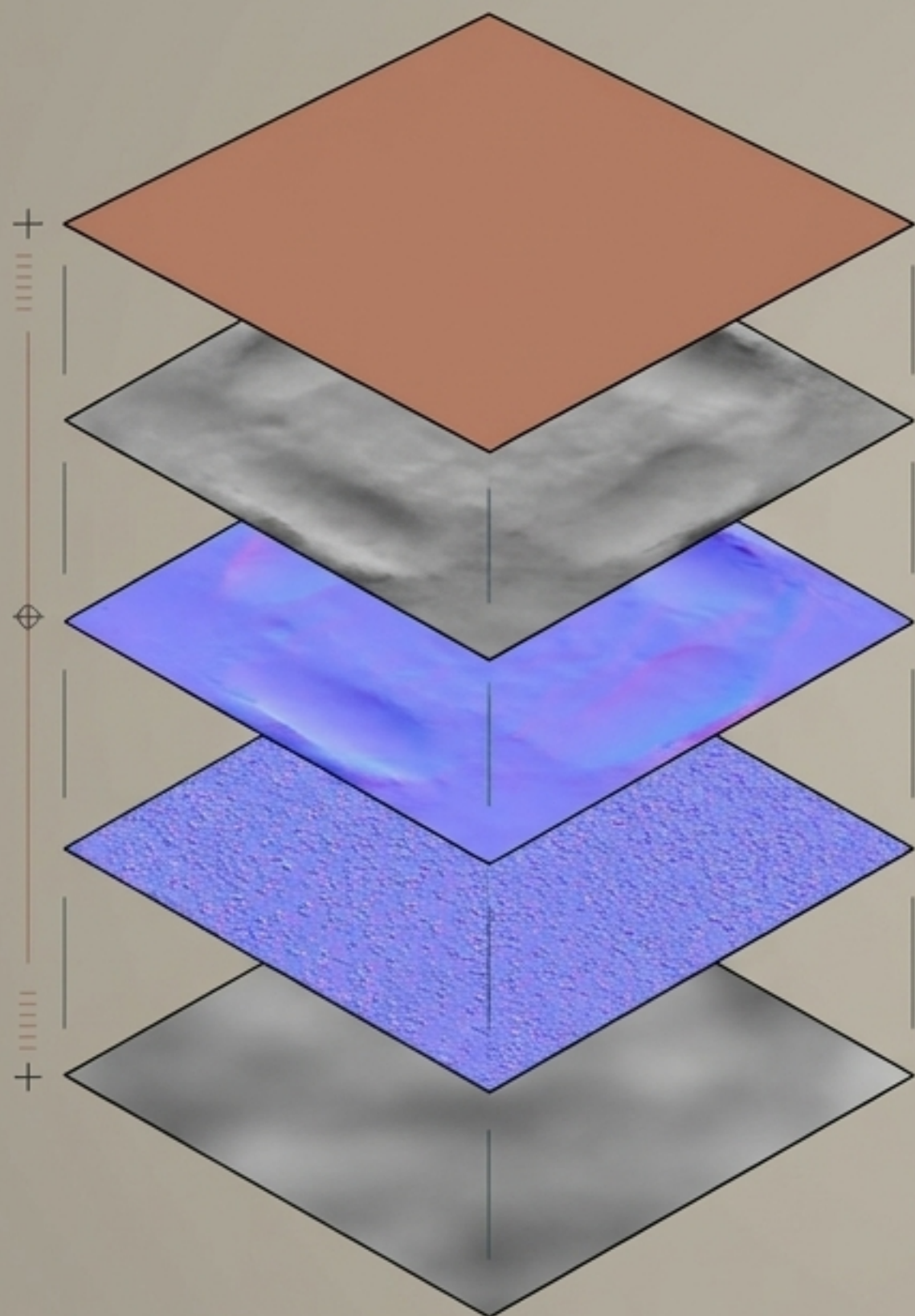
3. Rough / Porous
Clay: Cavity-heavy
for material
variations.

4. Compressed Clay:
Broad, flattened
manual pressure
simulation.

5. Rolled Clay:
Subtle directional
leveling.

(List continues sequentially through 10 in standard documentation)

The Digital Sculptor's Spec Sheet



Albedo / Base Color: [sRGB]
Intrinsic color tinting only. No baked shadows.

T_Clay_MesoRoughness_R: [Non-Color] 1024x1024.
Modulates matte finish between crevices and flats. 2-4x repeat.

T_Clay_MesoSculpt_N: [Non-Color] 2048x2048.
Encodes broad manual tool marks. 2-4x repeat.

T_Clay_MicroGrain_N: [Non-Color] 2048x2048.
Encodes fine starch grain. High frequency, 16-32x repeat.

T_Clay_MottleMask_M: [Non-Color] 1024x1024.
Low frequency soft color shift.

The Digital Sculptor's Spec Sheet

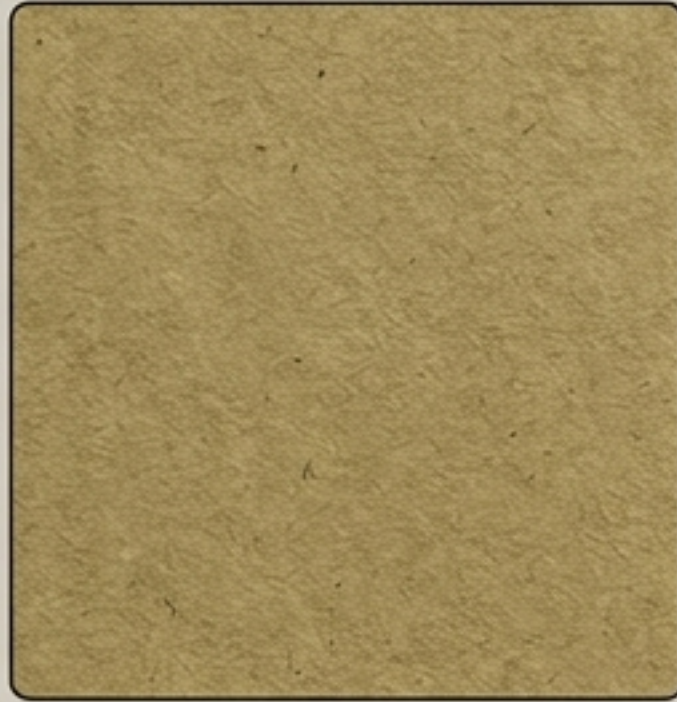
Secondary Materials



Felt / Woven Textile:

BSDF Override:
BSDF Override: Increase
Sheen weight to 0.50.
Apply woven normal map.

Secondary Materials



Paper / Cardboard:

BSDF Override:
Subsurface 0.000,
highest roughness,
macro-bump input.

Secondary Materials



Matte Rubber / Craft
Foam:

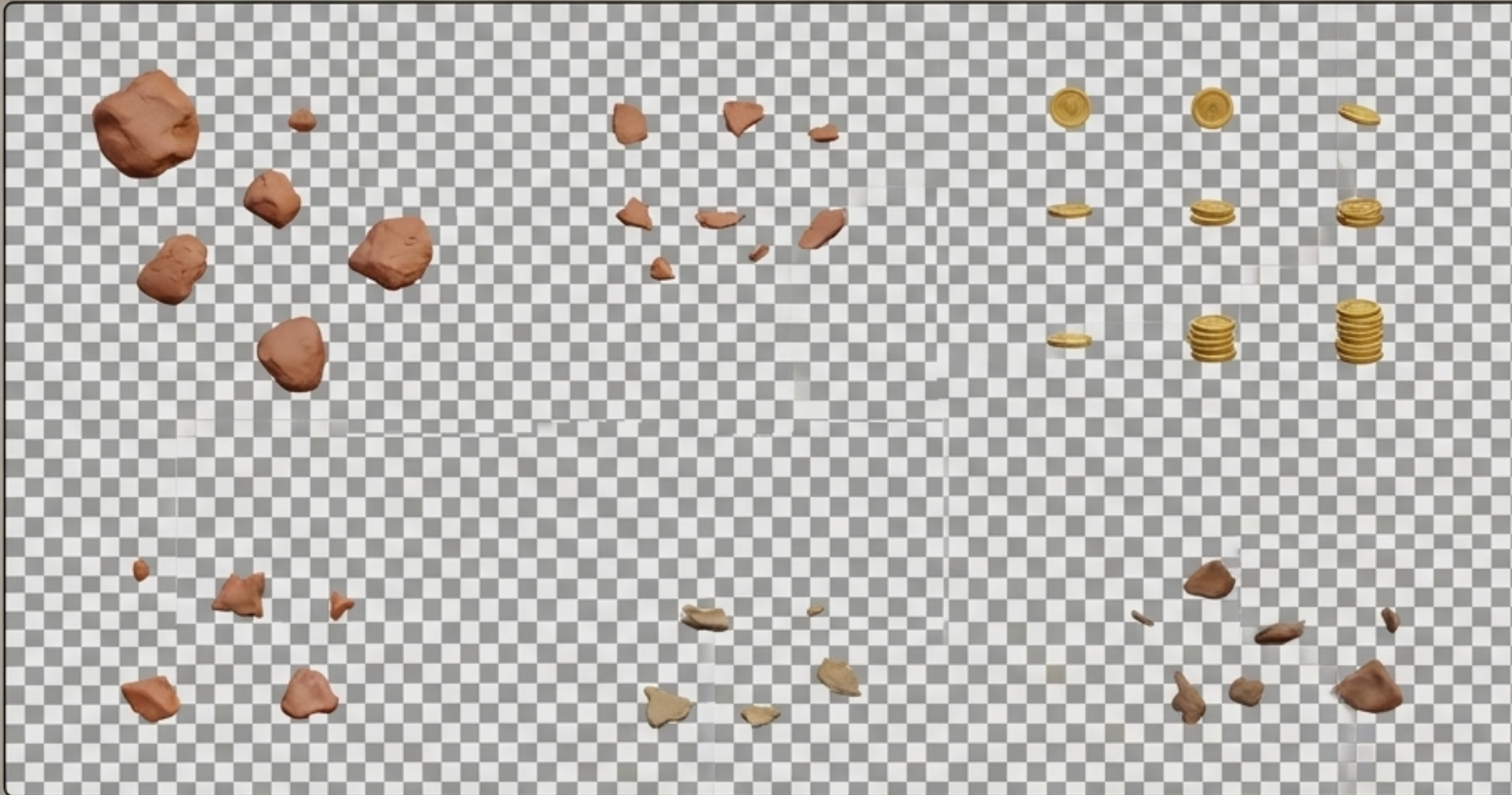
BSDF Override:
Subsurface 0.050, lower
roughness (0.500),
zero micro-grain.

Phase 2 Utility Masks Queue

- Broad non-repetitive macro color variation.
- Broad non-repetitive roughness variation.
- Triplanar-friendly color breakup.

PHASE 4 & 5 – IMPERFECTION & DECAL SHEETS

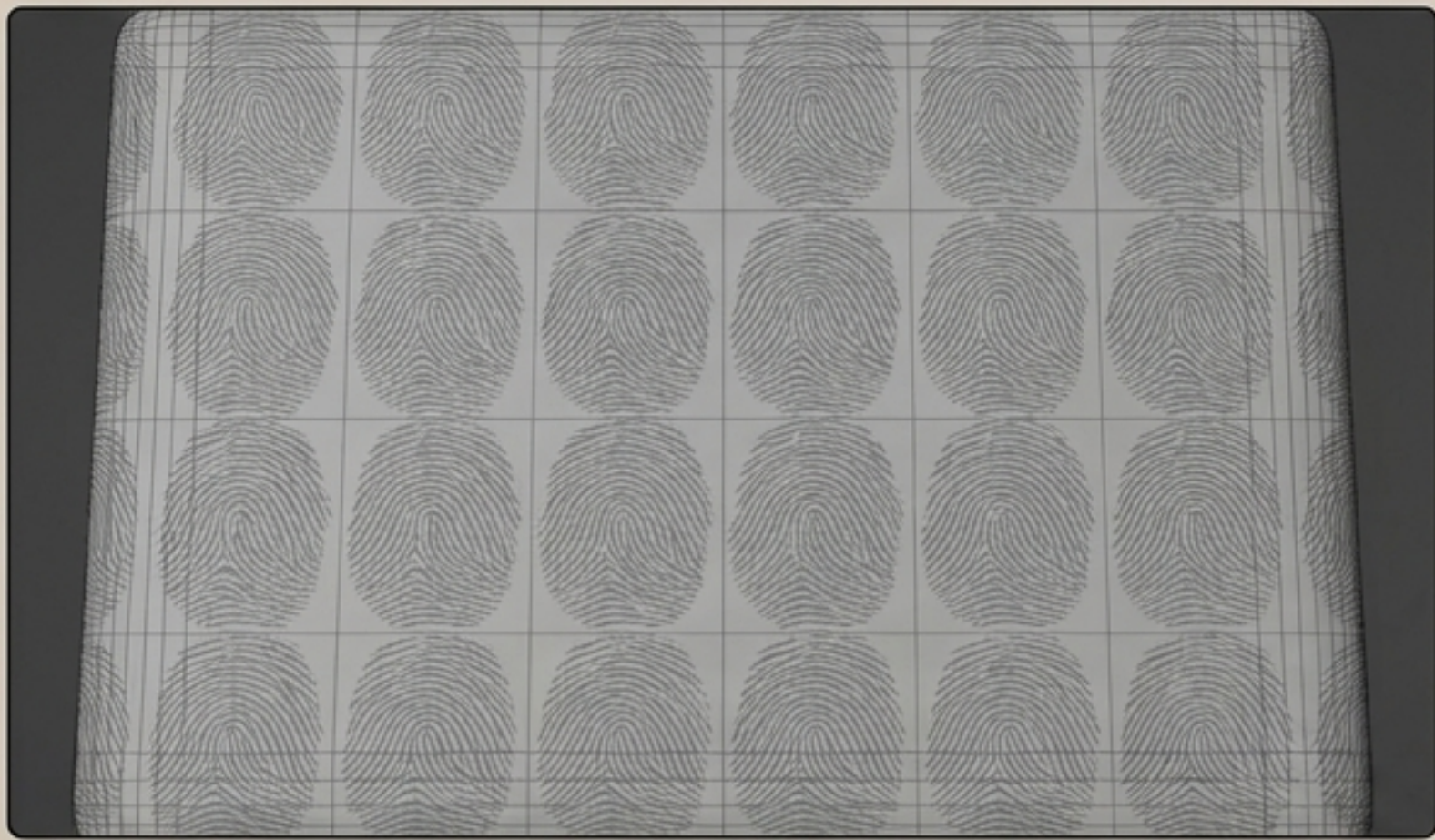
Formatting Rules: Isolate elements clearly. Use transparent backgrounds. Edges must be readable for extraction. Do NOT merge into noisy compositions.



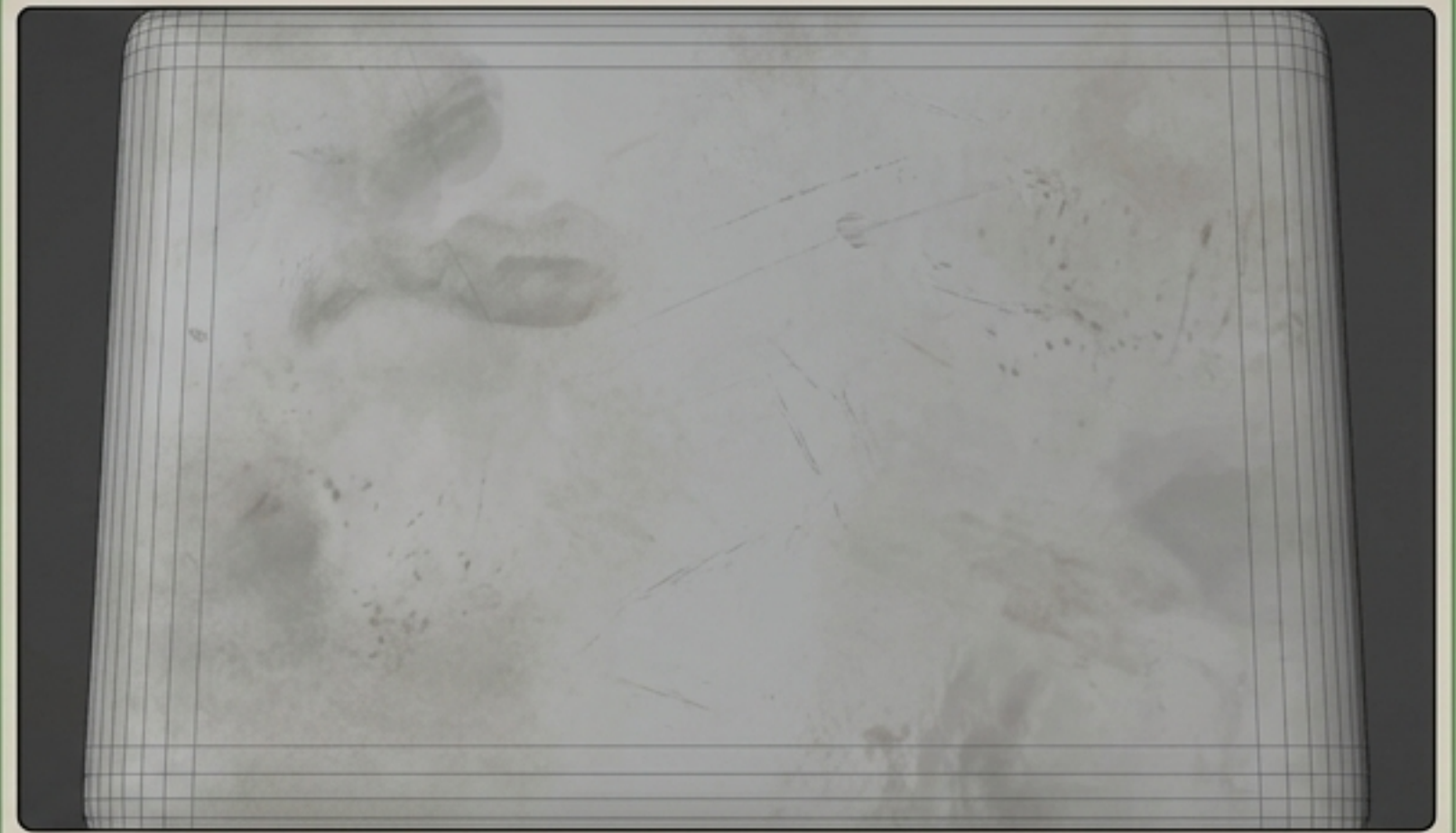
Asset Queue (Examples)

- 30. Clay dents & 31. Clay bumps
- 33. Thumb presses & 34. Spatula marks
- 39. Clay crumbs & 40. Clay flakes
- 43. Patch repair elements
- 45. DIY stop-motion diorama debris

DON'T (Bad Execution)



DO (Good Execution)



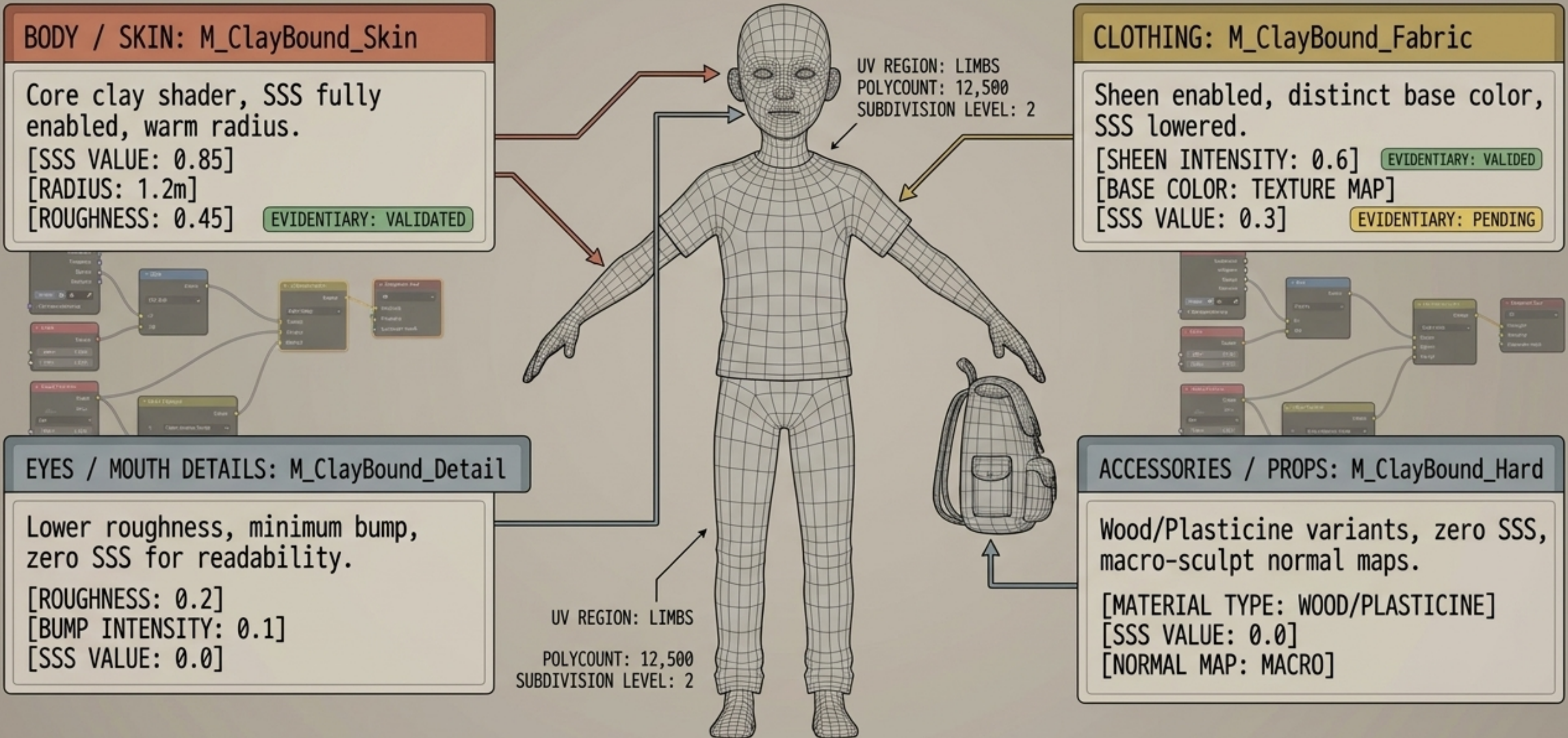
The Anti-Repetition Formula

Mechanism Specification

- **[INPUT]** - Input: Object Info Node (Random seed).
- **[MATH]** - Math: Multiply by scale scalar, feed into Combine XYZ.
- **[APPLICATION]** - Application: Drives UV rotation/offset for duplicated instances.
- **[COLOR SHIFT]** - Color Shift: Soft low-frequency mapping node (T_Clay_MottleMask_M) mixed with Base Color via Soft Light blend (Factor: 0.10 - 0.20).

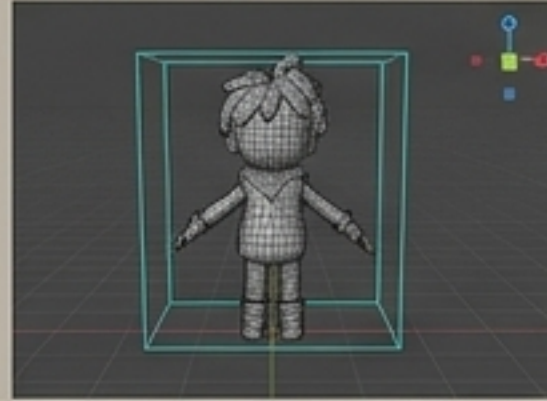


PHASE 6 – MATERIAL ASSIGNMENT MAP



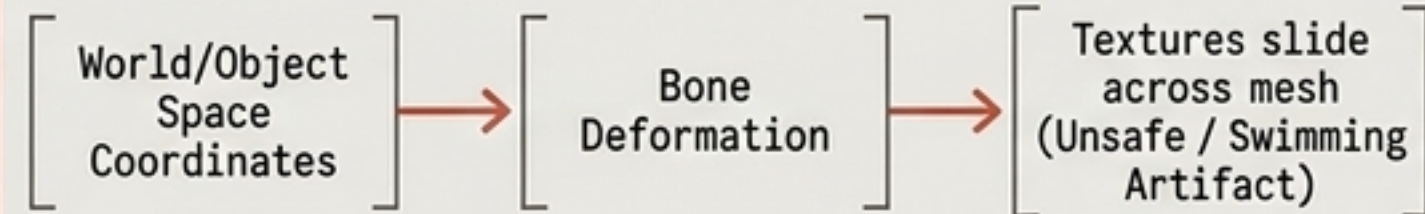
THE DIGITAL SCULPTOR'S SPEC SHEET

ANTI-SWIMMING ARCHITECTURE



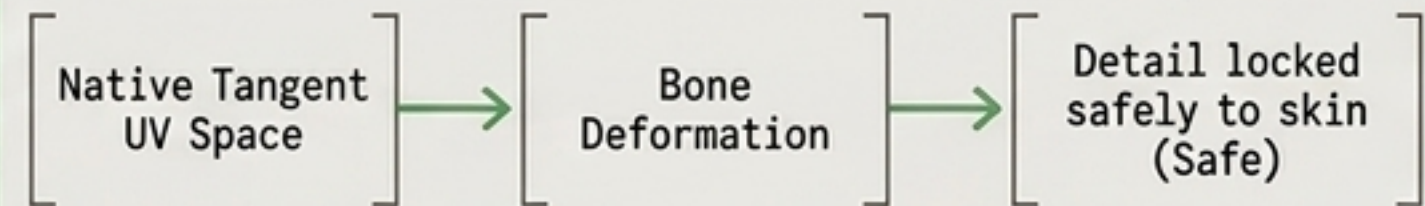
TOP PATH: (UNSAFE) WORLD/OBJECT SPACE COORDS

evidentary stats



BOTTOM PATH: (SAFE) NATIVE TANGENT UV SPACE

evidentary safe



STOP-MOTION JITTER FORMULA

evidentary safe

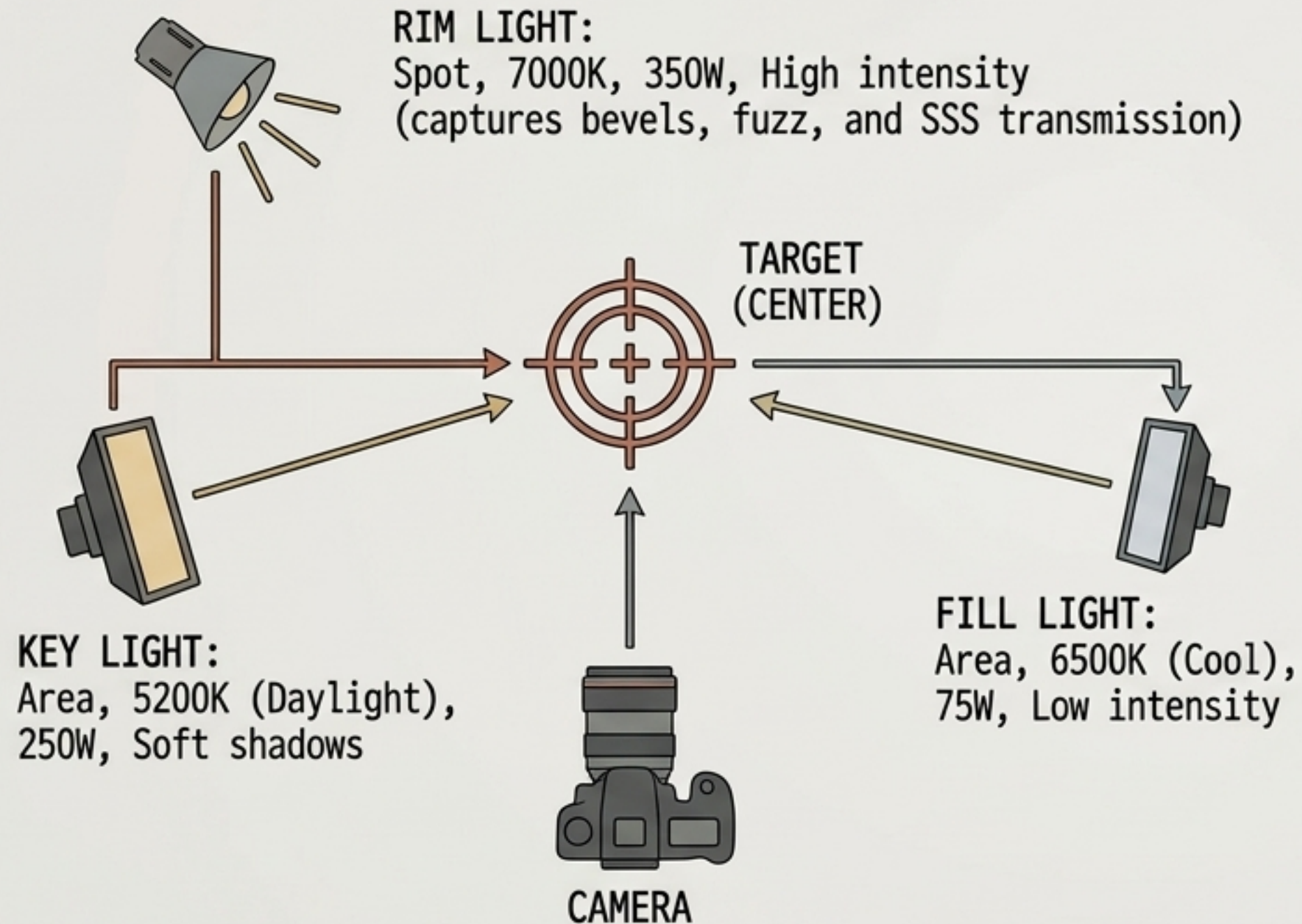
To achieve frame-quantized surface re-sculpting without swimming, step the procedural offsets using frame drivers:

$$\text{Offset_X} = \text{floor}(\text{Frame} / N) * C$$

(N = Frame step interval, C = Spatial step offset)

PHASE 7 – LIGHTING & CAMERA SPECIFICATIONS

STUDIO LIGHTING SCHEMATIC (TOP-DOWN VIEW)



CAMERA & COLOR SPEC BOX

- Focal Length: 85mm (DoF f/4.0 - f/5.6)
- Color Management: AgX View Transform
- Look: Medium High Contrast (Prevents SSS blowout while retaining rich color saturation)



THE DIGITAL SCULPTOR'S SPEC SHEET

DELIVERABLE STRUCTURE (FILE TREE)

/KFB_CLAY_MATERIAL_SYSTEM/

/textures/

clay_smooth_N.exr

clay_fine_R.png

/blender/

KFB_Clay_Master.blend

/specs/

MATERIAL_SPEC.md

(Action item: Ctrl+A to normalize scale to 1.0 before appending)

QA ACCEPTANCE CHECKLIST

- ☐ Does the silhouette lack sharp 90° polygonal edges?
- ☐ Is the surface velvet-matte with zero glossy plastic reflections?
- ☐ Does subsurface light scatter warmly along thin extremities?
- ☐ Are high-frequency repetitive fingerprints successfully avoided?
- ☐ Do textures stick to the mesh during bone deformation (no swimming)?
- ☐ Is the original KFB character identity immediately recognizable?